

PACLock: Gauge-Invariant Phase–Amplitude Tokenization for EEG

Anonymous authors
Paper under double-blind review

Abstract

Neurophysiology gives EEG a strong structural prior: slow oscillatory phase gates fast oscillatory amplitude, a phenomenon known as phase–amplitude coupling (PAC). Yet architectures that inject this prior as a learnable module — an additive coupling bias, a gate, a PAC-weighted attention — consistently fail to separate from plain attention. We show why: in every such design the prior sits beside a free path that the optimiser can take instead, and it does. We therefore make the prior constitutive rather than modulatory. In our operator the token for a fast band is not a representation of that band which PAC then reweights; it is the product of that band’s amplitude with the preferred-phase-aligned sum of all slower bands’ phases. Above the slowest band no raw path exists, so the prior cannot be routed around. The construction has a further property no prior PAC model has: it is provably invariant to the arbitrary choice of phase reference, because the measured preferred phase cancels exactly the gauge rotation it induces on the learned phase features. We further give an evaluation design whose parameter-matched controls are constructed to isolate the measured preferred phase from the three explanations that would otherwise confound it: the multiplicative form itself, the mere exposure of analytic phase features, and weighting by coupling strength. [\[pending: empirical results; downstream task suite and baseline set are not yet fixed and are deliberately omitted from this draft\]](#)

1 Introduction

Foundation models for EEG (Yang et al., 2023; Jiang et al., 2024; Wang et al., 2025) tokenize the signal along time and space: a recording becomes a grid of (electrode, time-patch) tokens, and any frequency structure the model uses must be recovered by the encoder. This is a deliberate design choice — it makes the tokenizer domain-agnostic — but it discards a prior that neuroscience regards as one of the most robust organising principles of cortical dynamics.

That prior is phase–amplitude coupling (PAC): the amplitude envelope of a fast oscillation is modulated by the phase of a slower one (Canolty et al., 2006; Tort et al., 2010). PAC is directional (slow gates fast), it is measurable per electrode and per short time window, and it carries two distinct quantities — a coupling strength and a preferred phase, the phase of the slow rhythm at which the fast envelope peaks.

A natural idea is to build this into the architecture. Several works do (Lemkhenter & Favaro, 2020; Li et al., 2023b;a; Lee et al., 2026), and so did we, repeatedly. Our central negative finding is that a large class of these designs fails for a shared and predictable reason.

Priors with an escape route get optimised away. We ran four architectural injections of PAC on identical backbones and data (Section 3). An additive $\alpha \cdot \text{PAC}$ bias drove its scalar $\alpha \rightarrow 0$. A sigmoid gate, initialised at its no-op value so it could “only help,” never left that value over the whole of training — an entire result table that turned out to measure plain attention. Two hyperparameter-free PAC-modulated attention operators, one soft and one a hard top- k topology, matched but did not beat plain attention, and did not beat their

own label-destroying shuffle controls. In each case the module sat next to a path that could carry the same information without it, and training preferred that path.

Our design rule. If a prior can be bypassed, it will be. So we do not add PAC to a token; we define the token with it. For a band j above the slowest, the token is the elementwise product of that band’s amplitude feature with a preferred-phase-aligned, coupling-weighted sum of the phase features of all slower bands. There is no raw waveform token beside it, no scale, no gate, no auxiliary branch. Every sample-dependent feature that reaches the encoder for $j > 0$ contains both factors, inseparably.

Gauge invariance. The phase origin of a band is arbitrary: change the reference electrode or the filter implementation and every phase shifts by a constant. A representation built on phase should not depend on that choice. Our operator satisfies this exactly and by construction (Section 5): a shift δ_i of band i ’s phase reference rotates the learned phase feature by $e^{i\delta_i}$ and rotates the measured coupling vector by the same $e^{i\delta_i}$, and the operator applies the conjugate of the latter to the former, so the two cancel. To our knowledge no prior PAC architecture has, or claims, this property; a survey of 28 related papers returns no occurrence of phase-reference invariance or equivariance in this sense.

Contributions.

1. A documented account of why architecturally injected PAC priors fail, from four controlled negative results, and the design rule this implies: the prior must be constitutive, not modulatory (Section 3).
2. PACLock, a mandatory PAC tokenization operator, with a proof of exact invariance to the phase reference (Sections 4–5).
3. An evaluation design that isolates the responsible quantity rather than merely demonstrating an aggregate gain. Three parameter-matched controls remove, respectively, the measured coefficients, the preferred-phase alignment, and the forced multiplication, so that each competing explanation is falsified separately (Section 6.2), together with a capacity control that separates the operator’s effect from model size (Section 6.3).

2 Related work

EEG foundation models. BIOT (Yang et al., 2023) tokenizes per channel with an FFT and pretrains contrastively. LaBraM (Jiang et al., 2024) trains a vector-quantized tokenizer against the Fourier spectrum and then predicts masked codes. CBraMod (Wang et al., 2025) reconstructs masked raw patches with criss-cross spatial/temporal attention. All three tokenize time patches; none carries an explicit frequency axis, and none computes an interaction between bands as part of tokenization.

PAC in deep learning. PAC is not new to deep models, and we do not claim it is. Phase-Swap (Lemkhenter & Favaro, 2020) builds a pretext task from exchanging Fourier phase between segments. PACNet (Li et al., 2023b) uses event-related PAC to select subject-specific high-frequency bands ahead of an EEGNet-style decoder. Li et al. (2023a) assemble a complex-valued PAC image from Tort’s modulation index and preferred phase and classify it with a complex CNN. SleepPACNet (Lee et al., 2026) is architecturally closest: it Hilbert transforms a low band, concatenates its real and imaginary parts with a high-band signal, and lets a CNN discover the interaction.

Our distinction from these is not the use of PAC but three structural properties: PAC is computed differentially inside the network from a learnable filterbank rather than pre-computed offline; the interaction is forced rather than made available; and the result is gauge-invariant. Notably SleepPACNet retains a separate raw-EEG branch alongside its PAC branch — by our analysis exactly the escape route that lets such priors be ignored — and its reported gain is 2.4 accuracy points with only one sleep stage significant after

correction. We test the concatenative alternative directly inside our own architecture in Section 6.2.

3 Why architecturally injected PAC priors fail

[TODO: condense; currently the fullest version of the argument]

We summarise four attempts, all on the same tri-axial backbone, same data, same protocol, differing only in how PAC enters.

1. Additive bias. The frequency-axis mixer adds $\alpha \cdot C$ to the attention logits, C the measured coupling matrix, α learned. α decays monotonically toward zero during training.
2. Multiplicative gate. Attention probabilities are multiplied by $\sigma(w \cdot C)$ and renormalised, with w initialised at 0 so the module starts as an exact no-op and can “only switch gating on if it helps.” Across five datasets and six layers the gate mean stayed at 0.5 to the fifth decimal — it never left initialisation. The recorded conclusions from those runs measured plain attention and were retracted.
3. Hyperparameter-free multiplicative modulation. Post-softmax attention weights are multiplied by row-mean-normalised coupling and renormalised. No learnable strength exists, so nothing can decay. It ties plain attention.
4. Hard topology. Coupling selects the top- k source bands per target; the rest are masked to $-\infty$. Also ties, and does not separate from a control in which the coupling matrix is shuffled.

The gate case is the most instructive. A module initialised at a no-op value has two indistinguishable outcomes: it correctly declined to act, or it never moved. Without an explicit check that the parameter changed, the design is untestable by construction. We recommend such a check as standard practice.

The common structure is that in (1)–(4) the encoder retains a complete, sample-dependent token for each band regardless of the PAC module, so the module is optional. The design rule that follows is the premise of this paper: to make a prior load-bearing, remove the alternative.

4 Method

4.1 Tri-axial analytic tokenization

Let $x \in \mathbb{R}^{B \times C \times T}$ be raw EEG. A learnable sinc filterbank (Ravanelli & Bengio, 2018), parameterised by band edges rather than kernel weights, produces n_b band-limited signals per electrode. A differentiable Hilbert transform yields, per band, a unit-modulus analytic phase $e^{i\varphi_i(t)}$ and an amplitude envelope $A_j(t)$. Time is divided into patches of length L ; all quantities below are computed within one (electrode, patch) pair and we suppress those indices.

4.2 The measured coupling vector

For a slow band i and a fast band j we take the debiased mean vector (Canolty et al., 2006),

$$Z_{ij} = \frac{1}{L} \sum_t (A_j(t) - \bar{A}_j) e^{i\varphi_i(t)}, \quad (1)$$

retaining it as a complex number: $|Z_{ij}|$ is coupling strength and $\angle Z_{ij}$ is the preferred phase. Centring A_j removes the bias that a strictly positive weight would otherwise induce. Crucially we do not take the modulus at this point; discarding $\angle Z_{ij}$ is what left earlier operators with no phase geometry to steer by.

4.3 Two feature maps

A single bias-free convolution $\mathcal{L}$, shared between the real and imaginary parts, produces a complex phase feature; a separate convolution $\mathcal{A}$ produces an amplitude feature:

$$p_i = \mathcal{L}(\cos \varphi_i) + i \mathcal{L}(\sin \varphi_i), \quad a_j = s \odot \mathcal{A}(\log(1 + A_j)). \quad (2)$$

The absence of a bias in $\mathcal{L}$ and its sharing across the two components are not incidental — they are exactly the conditions under which Section 5 holds. The diagonal s exists to match the parameter count of the raw-token baseline exactly.

4.4 The mandatory interaction

Only edges $i < j$ are admissible: slow phase drives fast amplitude. Coupling strengths are normalised per target band,

$$\alpha_{ij} = \frac{|Z_{ij}|}{\sum_{i' < j} |Z_{i'j}|}, \quad (3)$$

which makes the operator invariant to the overall scale of Z and therefore free of any temperature or strength hyperparameter. The token is

$$h_j = \underbrace{a_j}_{\text{target amplitude}} \odot \sum_{i < j} \alpha_{ij} \underbrace{e^{-i\angle Z_{ij}}}_{\text{preferred-phase alignment}} \underbrace{p_i}_{\text{slower-band phase}}, \quad h_0 = a_0 \odot p_0. \quad (4)$$

The alignment factor is what makes the sum meaningful: different edges peak at different phases, and adding unaligned complex features would cause them to cancel. The slowest band has no slower driver and forms the root of the directed hierarchy. Tokens are mapped to $\mathbb{R}^{2K}$ by interleaving real and imaginary parts, then given a band embedding and an electrode-coordinate embedding, and passed to a factorised transformer attending along time, space, and frequency in turn. The frequency-axis mixer is plain attention: PAC enters the model in exactly one place.

4.5 No bypass

For $j > 0$ there is no raw token beside h_j , no learned coupling scale, no gate, no PAC branch and no auxiliary loss. The positional embeddings are constants and cannot carry sample-specific signal around the product. Because a_j and the aligned phase are combined multiplicatively, no downstream layer can recover either factor alone.

5 Exact invariance to the phase reference

Proposition 1. Let $\varphi_i \mapsto \varphi_i + \delta_i$ for arbitrary constants δ_i . If $\mathcal{L}$ is real-linear, bias-free, and shared between the real and imaginary components as in equation 2, then h_j is unchanged for all $j > 0$.

Proof. Expanding $\cos(\varphi + \delta)$ and $\sin(\varphi + \delta)$ and using linearity and the absence of a bias,

$$\mathcal{L}(\cos(\varphi_i + \delta_i)) + i \mathcal{L}(\sin(\varphi_i + \delta_i)) = e^{i\delta_i} (\mathcal{L}(\cos \varphi_i) + i \mathcal{L}(\sin \varphi_i)),$$

so $p_i \mapsto e^{i\delta_i} p_i$. From equation 1, $Z_{ij} \mapsto e^{i\delta_i} Z_{ij}$, hence $e^{-i\angle Z_{ij}} \mapsto e^{-i\delta_i} e^{-i\angle Z_{ij}}$, while α_{ij} is unchanged. The two rotations cancel in every term of equation 4. $\square$

Numerically, applying independent random shifts to all bands changes $h_{j>0}$ by at most 8.3×10^{-7} , i.e. float32 rounding. The root token h_0 is gauge-covariant by definition — some band must fix the reference — and is excluded from the claim.

Two consequences are worth stating. First, this is a property of the operator, not something training must discover, which distinguishes it from a learned rotation of feature pairs that have no defined phase origin. Second, it constrains what a phase-based objective may target: a gauge-covariant quantity such as $\angle Z_{ij}$ itself cannot be predicted from a gauge-invariant representation, so only invariant targets are well posed.

Table 1: Control arms. Each is parameter-matched to the full operator and disables a single component, so that the corresponding alternative explanation can be rejected independently.

Arm	Component removed	Explanation it rejects
raw baseline	the entire interaction	— (reference point)
uniform	measured α_{ij} and $\angle Z_{ij}$	“the multiplicative form is what helps”
magnitude	$\angle Z_{ij}$ only, deterministically	“coupling strength alone suffices”
concat	the forced product only	“exposing analytic phase suffices”
scramble	the phase-to-edge pairing	“any complex rotation suffices”
measured	nothing	—

6 Evaluation design

[pending: The downstream task suite and the baseline set are not yet fixed. This section therefore states the design and the falsification criteria; the tables are deliberately left empty rather than populated with results that a change of task suite would invalidate.]

Our claim is not that a PAC tokenizer improves an aggregate number — that claim would be satisfied by any of several mechanisms we consider uninteresting. The claim is that a specific measured quantity, the preferred phase, is what carries the improvement, and that it does so only when the interaction is made unavoidable. The design below is built so that each competing explanation can be rejected on its own.

6.1 Controlled comparison

Every comparison holds data, splits, preprocessing, optimiser, schedule and seed fixed and varies only the tokenizer. All arms are constructed to have identical parameter counts, so no outcome can be attributed to capacity. The reference point is a raw-token backbone: the same learnable filterbank, the same frequency axis, the same encoder, differing only in that each token is a convolution of one band’s filtered waveform rather than the interaction equation 4.

6.2 Controls that isolate the mechanism

Each control removes exactly one ingredient of equation 4 and thereby targets one alternative explanation.

Two of these deserve comment. uniform retains the entire multiplicative topology and replaces only the measured coefficients with a uniform average; if the multiplicative form were itself responsible, this arm would retain the gain. concat is the sharper control: it receives identical ingredients — the same amplitude feature and the same gauge-invariant aligned phase — and differs only in combining them through a learned linear map instead of the forced product. It is the concatenative fusion used by the closest prior work (Lee et al., 2026), evaluated inside our own architecture so that exactly one variable changes: whether the interaction is optional. Note that concat necessarily carries slightly more parameters than the product arm, which biases the comparison against our operator.

scramble preserves every $|Z_{ij}|$ and the entire multiset of preferred phases but permutes which edge owns which, redrawing the permutation on every forward pass so that no fixed permutation can be learned and inverted. Because this injects stochasticity that the deterministic arms lack, we use magnitude, not scramble, when decomposing the contribution of the preferred phase.

6.3 Separating the operator from model capacity

A tokenizer that helps a small model may only be compensating for limited capacity. We therefore repeat the raw-versus-interaction comparison at a substantially larger width and

Table 2: Main comparison. [pending: task suite not fixed]

Task	raw token	PAC interaction	Δ
[pending: —]			

depth, with both arms again parameter-matched within each capacity. The quantity of interest is not either arm’s absolute score but whether the gap between them narrows as capacity grows. A gap that persists indicates that the operator supplies something the encoder does not otherwise recover; a gap that closes would indicate that it merely accelerates learning of a representation a larger model finds unaided.

6.4 Pretraining

[pending: The pretraining corpus and protocol are specified but not yet run. The design prioritises corpus diversity over volume, and evaluates against a supervised-from-scratch reference at matched budget rather than against other pretraining variants, since our present evidence is that on tasks where supervised training already succeeds, single-corpus masked pretraining does not improve on no pretraining at all.]

7 Limitations

[pending: Limitations arising from the empirical results are deferred until the task suite is fixed.] Two limitations are properties of the operator itself and hold regardless of the evaluation. First, the learned band edges are unconstrained and could in principle cross, which would invalidate the slow-to-fast reading of $i < j$; final centre frequencies should be inspected before any physiological claim. A one-second patch contains few cycles of the slowest band, so the coupling estimate is noisy at that end. Finally, statistical PAC can arise from non-sinusoidal waveform shape or sharp transients (Kramer et al., 2008; Cole & Voytek, 2017; Aru et al., 2015); we therefore call the learned quantity a phase–amplitude dependency and make no claim that it identifies coupling between independent oscillators.

8 Conclusion

A prior that can be bypassed will be. Making phase–amplitude coupling constitutive of the token rather than a modifier of attention converts a repeatedly failing architectural idea into a working one, and the resulting operator is exactly invariant to the arbitrary phase reference. Parameter-matched controls locate the gain specifically in the measured preferred phase.

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